

PVE Signal Engine — Setup

Honest, deterministic US-stock **prediction-market** intelligence engine (v2.1). LIVE only — no demo data.

Data reality

PVE.trade is a prediction-market API. It does **not** provide option chains, strikes, expirations, Greeks, IV, IV rank/surface, real OI, GEX, gamma flip, max pain, OPRA, options sweeps/blocks, real equity NBBO, or underlying equity OHLCV. This system never fabricates those values or relabels prediction-market data as options data.

1. Requirements

- Node.js 18+ (uses built-in node --test, ESM).
- No build step and no external runtime dependencies beyond Express.
- A PVE agent key for live data (without it the app still boots; all endpoints report the upstream error).

2. Install & run

```
npm install
cp .env.example .env # then edit .env
npm start # http://localhost:3000
npm test # 49 tests
```

3. Environment (.env)

```
PVE_AGENT_KEY=your_pve_agent_key # server-side only; never sent to the browser
DASHBOARD_PASSWORD=change_me # login password (timing-safe compare)
PORT=3000
# optional classifier universe (merged with built-in lists):
US_EQUITY_TICKERS=TSLA,AAPL,NVDA
US_EQUITY_ETFS=SPY,QQQ,IWM
EXTRA_DENY_TICKERS=MEME,FART
```

Security: the agent key lives only in server memory and is attached to upstream requests via the X-Agent-Key header. It is never exposed to client JavaScript. .gitignore prevents committing .env, node_modules/, logs, secrets, and credentials.

4. US-equity classifier

A server-side gate runs on every market before the engine sees it. It allows S&P 500 / major US equities / major ETFs and obvious US-stock event contracts, and hard-excludes crypto, politics, sports, weather, and similar categories. Anything ambiguous is rejected. Inspect decisions at GET /api/_classify.

5. Verify live US-equity coverage (once the key is enabled)

- Add PVE_AGENT_KEY to .env and restart.
- Log in, open **System Health**: PVE connection = reachable, Live key = yes, non-zero markets received / US-equity accepted.
- Open /api/_classify to see per-market ALLOW/REJECT reasons.
- Open **US Stock Signals** for ranked high-scoring US-equity/ETF prediction markets.

- Open **API / Data Monitor** for versions and received/accepted/rejectedd counts.

6. 403 / no-key handling

On HTTP 403 or a missing key the API returns `{ ok:false, error }` with no fabricated data, and the UI shows the precise message ("Insufficient live PVE data" / "PVE access unavailable — agent authorization required"). The application still starts and functions structurally.

Signal tiers (ranking, not win probabilities): 90–100 Exceptional · 80–89 Very Strong · 70–79 Strong · 60–69 Moderate · <60 Weak. A score is $|\text{net tilt}| \times \text{coverage}$ — never a probability of profit. See [UPGRADE.md](#) for the full report.